HcZZ ctag2d -- Results (Other_Higgs systematics completed, JES/JER, 2026-09-21)

Production: combine_run3_100150_ctag2d_rawsf_nozx_jecshifts_2026_09_21
hplusc_mva_4class_ctag2d_scored_v8 + hplusc_mva_4class_ctag2d_jecshifts_scored_v2,
[100,150] GeV, no-ZX (MC-only), 20 quantile bins, 23 nuisances. CMS_ctag2d uses
the OFFICIAL up_Total/down_Total, unmodified (L0 swap correction remains
reverted since 2026-09-15). Other_Higgs's lhe_pdf/scalevar_muR/scalevar_muF
coverage completed 2026-09-20 (previously excluded entirely; only its private
HPlusBottom sub-sample, ~4% of yield, is now masked). Rebuilt and verified
2026-09-21.

Yields [100,150] GeV

Process	Rate
Signal	0.053130
ggZZ	6.677889
qqZZ	36.318223
Other_Higgs	55.077609

$S/\sqrt{B}$ = 0.0054

Final limits (AsymptoticLimits, --run blind)

Quantile	r	kappa_c
2.5%	178.6641	33.63
16%	242.9138	44.79
50% (median)	**346.5000**	**62.75**
84%	510.8585	91.24
97.5%	730.3098	129.26

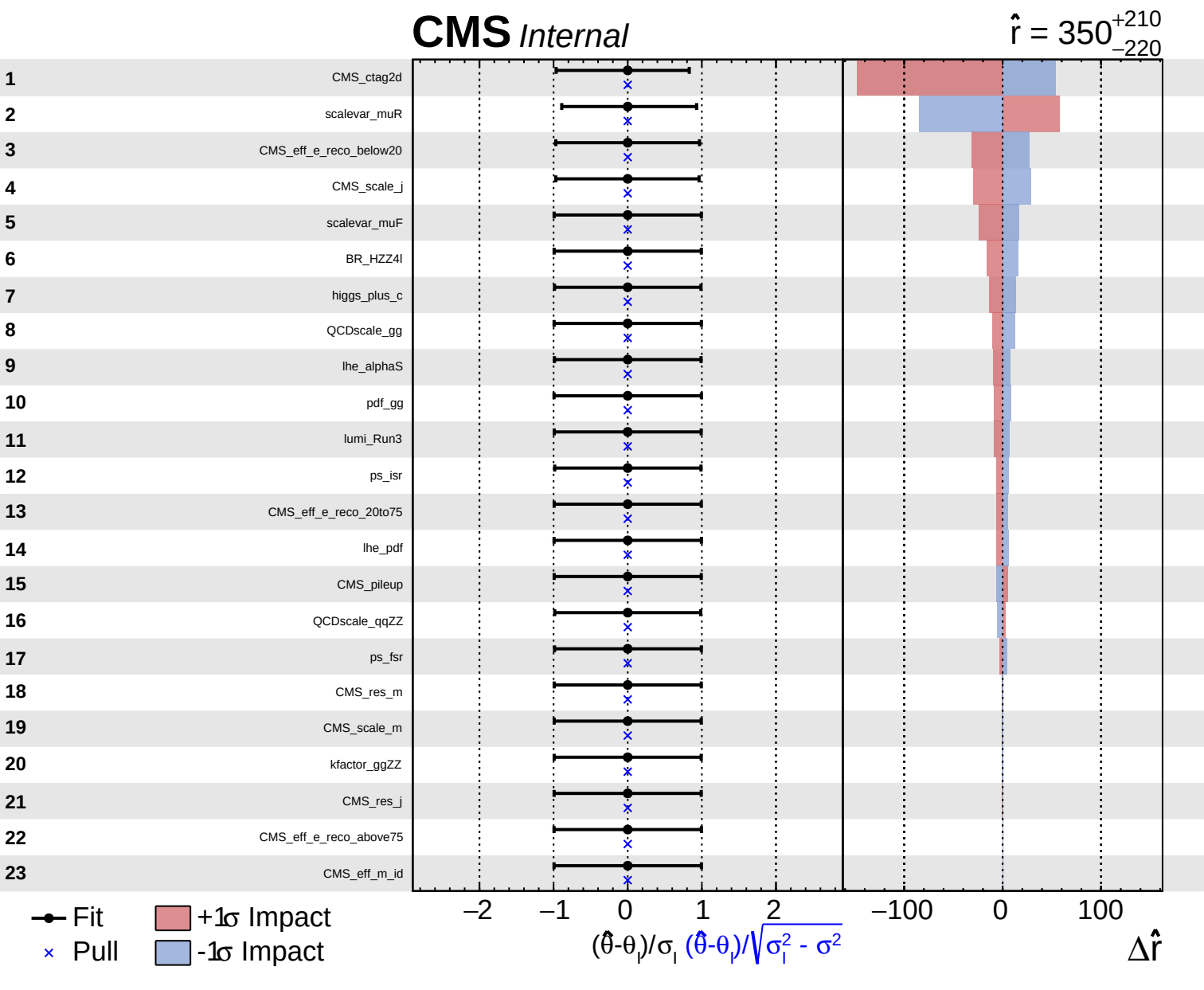
Stat-only (--freezeParameters allConstrainedNuisances): r=273.5/kappa_c=50.09 --
bit-identical to the pre-Other_Higgs-fix value (a pure systematics-side change
cannot move a stat-only fit). Full-syst median moved 339.5 (no JES/JER, post-fix)
-> 346.5 (this page, +JES/JER) -- +2.06% from JES/JER alone.

CMS_ctag2d is still #1 but no longer dominates by ~4x -- now only ~1.74x the
#2 nuisance (scalevar_muR, a real Other_Higgs shape for the first time,
impact 84.96 vs CMS_ctag2d's 147.68). CMS_scale_j ranks #5 (29.79).

Systematics: type, assigned value, and impact

Ranked by impact_r (combineTool.py -M Impacts, --expectSignal 346.5, official ctag2d SFs, 23 nuisances, Other_Higgs systematics completed). lnN rows show the assigned size per affected process; shape rows show the post-fit pull instead (all prefit, max |pull| ~1e-7, expected for a blind Asimov fit).

#	Systematic	Type	Value (lnN size / pull)	Impact
1	**CMS_ctag2d**	**shape**	**pull -0.00 (-0.966/+0.830)**	**147.68**
2	scalevar_muR	shape	pull -0.00 (-0.890/+0.930)	84.96
3	CMS_eff_e_reco_below20	shape	pull +0.00 (-0.970/+0.969)	31.33
4	CMS_scale_j	shape	pull +0.00 (-0.970/+0.963)	29.79
5	scalevar_muF	shape	pull +0.00 (-0.993/+0.993)	23.67
6	BR_HZZ4l	lnN	1.02 (Signal, Other_Higgs)	15.76
7	higgs_plus_c	shape	pull +0.00 (-0.987/+0.988)	13.92
8	QCDscale_gg	lnN	1.039 (Signal, ggZZ)	12.29
9	lhe_alphaS	shape	pull +0.00 (-0.989/+0.990)	9.89
10	pdf_gg	lnN	1.032 (Signal, ggZZ)	8.57
11	lumi_Run3	lnN	1.014 (Signal, ggZZ, qqZZ, Other_Higgs)	8.54
12	ps_isr	shape	pull +0.00 (-0.985/+0.989)	6.46
13	CMS_eff_e_reco_20to75	shape	pull +0.00 (-0.992/+0.992)	6.34
14	lhe_pdf	shape	pull -0.00 (-0.994/+0.994)	6.12
15	CMS_pileup	shape	pull -0.00 (-0.992/+0.992)	6.11
16	QCDscale_qqZZ	lnN	1.04 (qqZZ)	5.13
17	ps_fsr	shape	pull +0.00 (-0.987/+0.993)	4.85
18	CMS_res_m	shape	pull +0.00 (-0.992/+0.992)	0.98
19	CMS_scale_m	shape	pull +0.00 (-0.992/+0.992)	0.94
20	kfactor_ggZZ	lnN	1.10 (ggZZ)	0.86
21	CMS_res_j	shape	pull +0.00 (-0.993/+0.991)	0.26
22	CMS_eff_e_reco_above75	shape	pull +0.00 (-0.994/+0.994)	0.10
23	CMS_eff_m_id	shape	pull -0.00 (-0.994/+0.994)	0.02


CMS Internal
r̂ = 350 +210 −220
1 CMS_ctag2d
2 scalevar_muR
3 CMS_eff_e_reco_below20
4 CMS_scale_j
5 scalevar_muF
6 BR_HZZ4l
7 higgs_plus_c
8 QCDscale_gg
9 lhe_alphaS
10 pdf_gg
11 lumi_Run3
12 ps_isr
13 CMS_eff_e_reco_20to75
14 lhe_pdf
15 CMS_pileup
16 QCDscale_qqZZ
17 ps_fsr
18 CMS_res_m
19 CMS_scale_m
20 kfactor_ggZZ
21 CMS_res_j
22 CMS_eff_e_reco_above75
23 CMS_eff_m_id
−2 −1 0 1 2
−100 0 100
Fit
Pull
+1σ Impact
-1σ Impact
(θ̂-θ_I)/σ_I
(θ̂-θ_I)/√(σ_I² - σ²)
Δr̂

HcZZ -- with Z+X (current 4-era estimate) results, official SFs

Production: combine_run3_100150_ctag2d_rawsf_withzx_jecshifts_2026_09_21 [100,150] GeV, ctag2d scheme, 25 nuisances (18 shape/lnN + 4 JES/JER + ZX_norm/ZX_rate), current 4-era Z+X estimate via --zx-dir. Official ctag2d SFs (L0 swap correction reverted), Other_Higgs systematics completed 2026-09-20 (see no-ZX page). Note: the Z+X yield below (31.11) is smaller than a 2026-09-15 value (133.24) this doc previously carried -- the fake-rate production itself changed in between, unrelated to this update.

Yields in [100,150] GeV

Process	Yield
Signal	0.0531
ggZZ	6.6779
qqZZ	36.3182
Other_Higgs	55.0776
Z+X	31.1075
Total bkg	**129.1813**

$S/\sqrt{B}$ = 0.0047

Final limits (combine -M AsymptoticLimits -m 120 --run blind)

Quantile	r	kappa_c
2.5%	180.9844	34.03
16%	246.0685	45.33
50% (median)	**351.0000**	**63.53**
84%	519.5917	92.75
97.5%	745.3238	131.86

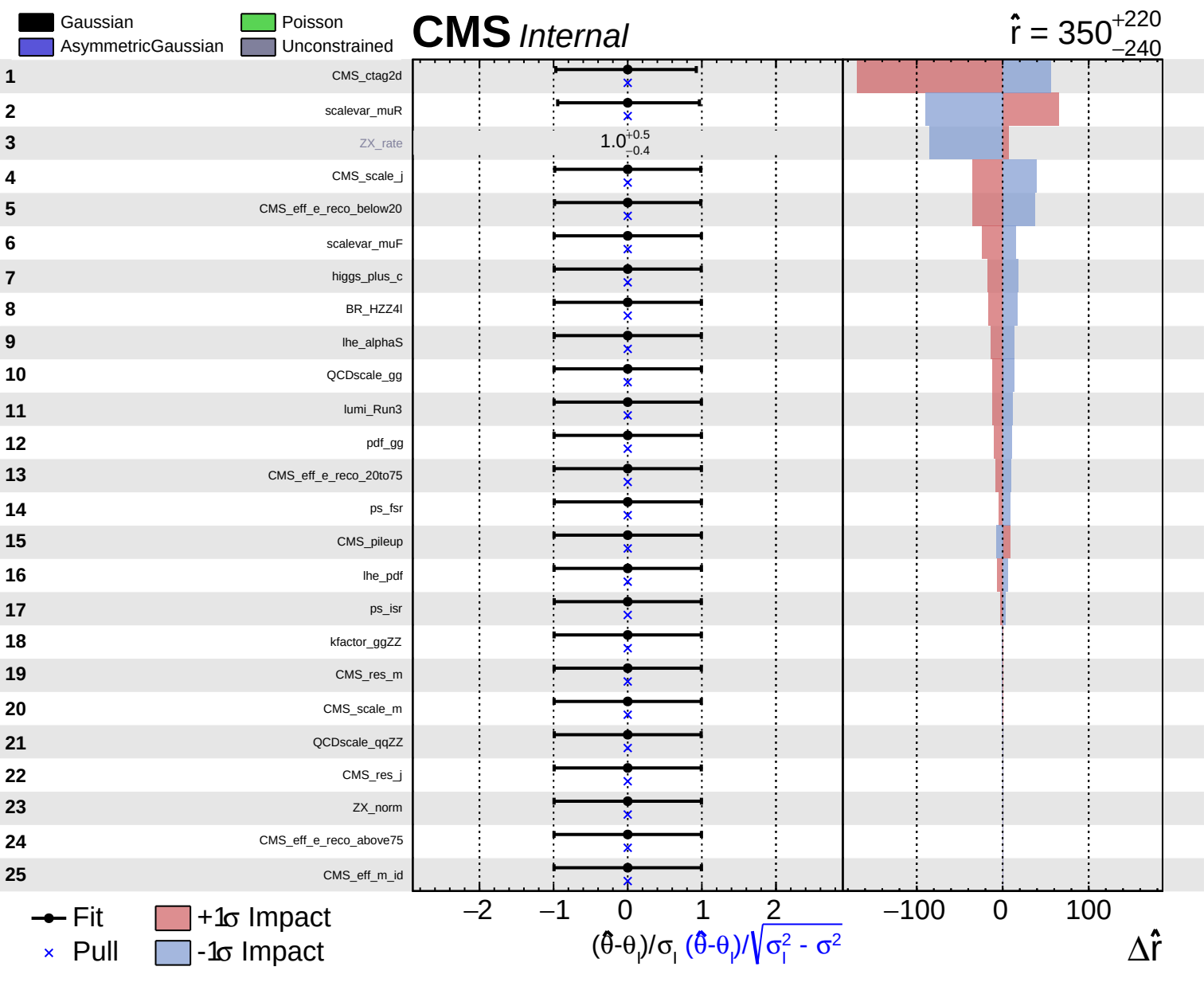
Comparison against the no-ZX headline

Configuration	r (median)	kappa_c
no-Z+X headline (23 syst, official SFs)	346.5	62.75
with-Z+X, current 4-era (25 syst, official SFs)	351.0	63.53

Systematics with Z+X: type, assigned value, and impact

Ranked by impact_r (combineTool.py -M Impacts, --expectSignal 351.0, official ctag2d SFs, current 4-era Z+X included, 25 nuisances incl. the free-floating ZX_rate). ZX_rate is now #3 (impact 84.88) -- CMS_ctag2d (169.10) and scalevar_muR (89.92) still rank #1/#2 ahead of it.

#	Systematic	Type	Value (lnN size / pull)	Impact
1	**CMS_ctag2d**	**shape**	**pull -0.00 (-0.970/+0.925)**	**169.10**
2	scalevar_muR	shape	pull +0.00 (-0.944/+0.967)	89.92
3	ZX_rate	rateParam	1.00 (+0.53/-0.43), free-floating	84.88
4	CMS_scale_j	shape	pull -0.00 (-0.985/+0.986)	39.39
5	CMS_eff_e_reco_below20	shape	pull +0.00 (-0.988/+0.986)	37.13
6	scalevar_muF	shape	pull +0.00 (-0.993/+0.994)	23.64
7	higgs_plus_c	shape	pull +0.00 (-0.992/+0.992)	18.40
8	BR_HZZ4l	lnN	1.02 (Signal, Other_Higgs)	17.48
9	lhe_alphaS	shape	pull +0.00 (-0.993/+0.993)	13.76
10	QCDscale_gg	lnN	1.039 (Signal, ggZZ)	13.38
11	lumi_Run3	lnN	1.014 (Signal, ggZZ, qqZZ, Other_Higgs, ZX)	11.64
12	pdf_gg	lnN	1.032 (Signal, ggZZ)	10.91
13	CMS_eff_e_reco_20to75	shape	pull -0.00 (-0.994/+0.994)	9.28
14	ps_fsr	shape	pull +0.00 (-0.992/+0.994)	8.94
15	CMS_pileup	shape	pull +0.00 (-0.994/+0.994)	8.65
16	lhe_pdf	shape	pull +0.00 (-0.994/+0.994)	5.89
17	ps_isr	shape	pull +0.00 (-0.992/+0.993)	3.24
18	kfactor_ggZZ	lnN	1.10 (ggZZ)	0.69
19	CMS_res_m	shape	pull +0.00 (-0.993/+0.993)	0.46
20	CMS_scale_m	shape	pull +0.00 (-0.993/+0.993)	0.32
21	QCDscale_qqZZ	lnN	1.04 (qqZZ)	0.17
22	CMS_res_j	shape	pull +0.00 (-0.993/+0.993)	0.17
23	ZX_norm	lnN	1.30 (ZX)	0.08
24	CMS_eff_e_reco_above75	shape	pull +0.00 (-0.994/+0.994)	0.06
25	CMS_eff_m_id	shape	pull +0.00 (-0.994/+0.994)	0.03


Gaussian
AsymmetricGaussian
Poisson
Unconstrained
CMS Internal
r̂ = 350 +220 −240
1 CMS_ctag2d
2 scalevar_muR
3 ZX_rate
1.0 +0.5 −0.4
4 CMS_scale_j
5 CMS_eff_e_reco_below20
6 scalevar_muF
7 higgs_plus_c
8 BR_HZZ4l
9 lhe_alphaS
10 QCDscale_gg
11 lumi_Run3
12 pdf_gg
13 CMS_eff_e_reco_20to75
14 ps_fsr
15 CMS_pileup
16 lhe_pdf
17 ps_isr
18 kfactor_ggZZ
19 CMS_res_m
20 CMS_scale_m
21 QCDscale_qqZZ
22 CMS_res_j
23 ZX_norm
24 CMS_eff_e_reco_above75
25 CMS_eff_m_id
−2
−1
0
1
2
−100
0
100
Fit
Pull
+1σ Impact
-1σ Impact
(θ̂-θ_I)/σ_I
(θ̂-θ_I)/√(σ_I² - σ²)
Δr̂

Stat-only limit: with vs. without Z+X (official SFs)

combine -M AsymptoticLimits --freezeParameters allConstrainedNuisances
(all constrained nuisances frozen; ZX_rate still floats since it is
unconstrained), same official-SF production, [100,150] GeV, both re-verified
2026-09-21. No-ZX stat-only is bit-identical to every prior rebuild (a pure
systematics-side fix cannot move it); with-ZX stat-only moved (283.5->289.5)
purely because the Z+X estimate itself changed (133.24 ->31.11 yield),
unrelated to this session's systematics work.

Quantile	r (no ZX)	kc (no ZX)	r (with ZX)	kc (with ZX)	dr	dr %
2.5%	141.0234	27.08	148.1426	28.32	+7.12	+5.0%
16%	190.7021	35.72	202.2560	37.73	+11.55	+6.1%
50% (median)	**273.5000**	**50.09**	**289.5000**	**52.87**	**+16.00**	**+5.9%**
84%	396.6906	71.45	421.6284	75.78	+24.94	+6.3%
97.5%	553.8622	98.69	590.0568	104.96	+36.19	+6.5%

Key point: stat-only, Z+X loosens the median limit by ~5.9%
(r=273.5→289.5) -- somewhat larger than the ~1.3% full-syst loosening
(r=346.5→351.0, official SFs). The full-syst effect is small because the
free-floating ZX_rate parameter absorbs most of Z+X's own normalization
freedom in the fit, leaving little room for the systematics side to add on
top of the smaller current Z+X yield's own statistical effect.